

AI Labs : AI Solutions - Ank Kumar

EAAP Enterprise AI Automation Platform

Amazon Bedrock Knowledge Base : BIAW4U4TIL · ap-southeast-2
Answer from documents : Claude Haiku 4.5 · au.anthropic.claude-haiku-4-5-20251001-v1:0
Design with the framework : Claude Sonnet 4.5 · au.anthropic.claude-sonnet-4-5-20250929-v1:0

QUERY

Answer from documents

Design with the framework

Applies the framework to a new problem or domain. Framework elements are cited; Claude's own proposals are tagged.

DOMAIN

healthcare

USE-CASE

Prescription Refill

Run query

New conversation

first text 4583 ms · total 63073 ms

Start from a known question

What are the core stages of the Enterprise AI Automation Framework?

How does AAIT implement AI-augmented incident triage?

What governance controls sit between an AI decision and an automated action?

DESIGN

Propose an AI Solution architecture using the Enterprise AI Automation Framework.

RESPONSE

1

 taken from your documents ·

proposed

 Claude's own design reasoning, not in the documents

AI Solution Architecture: Healthcare Prescription Refill Automation

Stage 1: Signal Ingestion

5

Objective: Capture prescription refill requests and contextual patient signals from multiple healthcare touchpoints.

Signal Sources proposed :

- Patient portal refill requests (web/mobile app submissions)
- Pharmacy system refill notifications (auto-refill programs, low-inventory alerts)
- Provider EHR systems (prescription expiration dates, treatment plan updates)
- Patient engagement platforms (SMS/email refill reminders with responses)
- Insurance eligibility feeds (formulary changes, prior authorization status)
- Medication adherence monitoring devices (smart pill bottles, connected inhalers)

Signal Attributes proposed :

- Patient identifiers, medication details (NDC, dosage, quantity)
- Prescription metadata (original date, refills remaining, prescriber)
- Refill timing (days since last fill, early refill flag)
- Clinical context flags (recent hospitalizations, lab results, active diagnoses)
- Insurance status and formulary tier

Ingestion Pattern 5 : Real-time event streams for patient-initiated requests; scheduled batch ingestion for proactive refill eligibility checks from pharmacy systems.

Stage 2: Knowledge Retrieval 5

Objective: Enrich refill requests with patient history, clinical guidelines, and regulatory context to inform reasoning.

Knowledge Sources proposed :

- Patient medication history (fill dates, adherence patterns, prior denials)
- Active prescription records (dosage changes, drug interactions, contraindications)
- Clinical decision support databases (evidence-based refill protocols, safety alerts)
- Insurance formulary and prior authorization rules
- Regulatory compliance knowledge base (controlled substance schedules, refill limits, prescriber DEA status)
- Provider preferences and standing refill protocols

Retrieval Strategy 5 : Vector search for similar patient cases and guideline matching; deterministic lookup for regulatory rules and active prescriptions; graph traversal for drug interaction checks.

Governance Note 1 5 : Controlled substance refills require deterministic validation of DEA schedules and days-supply limits in this stage, not deferred to LLM reasoning.

Stage 3: Prompt Construction 1 4

Objective: Assemble structured prompts that frame the refill decision with patient context, clinical rationale, and safety constraints.

Prompt Components proposed :

- Patient summary (age, active conditions, allergy list, recent hospitalizations)
- Medication request details (drug, dosage, quantity, refills remaining)
- Refill timing analysis (expected vs. actual request date, adherence score)
- Clinical flags (drug interactions, duplicate therapy, dosage concerns)
- Insurance context (formulary status, prior authorization requirement)
- Regulatory constraints (controlled substance rules, prescriber authority)
- Decision task: "Assess whether this refill should be auto-approved, requires prescriber review, or should be denied with specific rationale."

Prompt Engineering proposed: Few-shot examples of refill scenarios with reasoning chains; explicit instructions to cite safety concerns and regulatory barriers; constraint to recommend next action (approve, escalate, deny) with confidence score.

Stage 4: LLM / AI Reasoning 1 4

Objective: Generate clinical and operational reasoning about the refill request, surfacing risks, policy conflicts, and approval recommendations.

Reasoning Tasks proposed:

- Adherence pattern interpretation (early refill justified by travel, or misuse signal?)
- Clinical appropriateness (dosage consistent with diagnosis, no recent contraindications)
- Safety assessment (interaction risk with newly added medications)
- Policy compliance (insurance rules, prescriber authorization scope)
- Escalation triage (which requests need pharmacist or prescriber review vs. auto-approval)

Model Selection proposed: General-purpose LLM with healthcare fine-tuning for clinical reasoning; no real-time constraint if decision latency of 10-30 seconds acceptable for non-urgent refills.

Latency Consideration: Emergency refills (e.g., insulin, rescue inhalers) bypass LLM reasoning entirely and route to deterministic fast-path approval if basic safety checks pass proposed.

Stage 5: Structured Decision 1 4 5

Objective: Transform LLM reasoning into a standardized refill decision object with explicit action, rationale, and confidence.

Decision Schema proposed:

```
{
  "refill_action": ["auto_approve", "pharmacist_review", "prescriber_consult", "deny"],
  "confidence_score": 0.0-1.0,
  "clinical_rationale": "text summary",
  "safety_flags": ["interaction_risk", "early_refill", "dosage_concern", ...],
  "regulatory_barriers": ["controlled_substance_limit", "prescriber_authorization", ...],
  "recommended_next_steps": "text for patient/provider communication",
  "data_sources_cited": ["patient_history", "formulary_rules", "clinical_guideline_X"]
}
```

Deterministic Overrides proposed : Hard stops for expired prescriptions, invalid DEA numbers, or formulary non-coverage force "deny" or "prescriber_consult" regardless of LLM output.

Stage 6: Confidence / Governance 1 4 5

Objective: Apply risk-based governance rules to determine whether the refill decision can execute autonomously or requires human review.

Confidence Thresholds 1 5:

- **Auto-approve path:** Confidence ≥ 0.90 , no safety flags, routine maintenance medication, patient adherence history good, insurance pre-authorized proposed
- **Pharmacist review path:** Confidence 0.70-0.89, or any safety flag present, or early refill without clear justification proposed
- **Prescriber consult path:** Confidence < 0.70 , controlled substance, dosage change needed, or regulatory barrier proposed
- **Deny path:** Hard regulatory block (e.g., refills exhausted, prescription expired) proposed

Human Oversight Controls 1 5:

- All controlled substance refills require pharmacist verification before execution proposed
- Denials with patient dispute flag escalate to prescriber proposed
- Random audit sample (5%) of auto-approved refills reviewed weekly proposed

Audit Trail 5: Every decision logs patient ID, medication, LLM reasoning summary, confidence score, governance path taken, and approver identity (system or human) proposed.

Stage 7: Execution & Feedback 4 5

Execution Paths 1 4:

Automated Execution proposed:

- Transmit approval to pharmacy system (NCPDP SCRIPT standard)
- Update EHR refill status
- Send patient notification (app push, SMS, email)
- Log insurance claim initiation

Human Approval proposed:

- Route to pharmacist/prescriber work queue with decision summary and flagged concerns
- Capture human decision (approve, modify, deny) and free-text rationale
- Execute approved action via same pharmacy integration

Feedback Capture 5 proposed:

- Pharmacist/prescriber override decisions (approval reversed, denial overturned)
- Patient outcomes (medication picked up, abandoned, adverse event reported)

- Insurance adjudication results (claim paid, prior authorization triggered)
- Prescriber follow-up actions (dosage changed, medication switched)

Stage 8: Offline Improvement 4 5

Objective: Continuously refine reasoning quality, confidence calibration, and governance thresholds using execution outcomes.

Analysis Activities 5 proposed:

- **Confidence calibration:** Compare predicted confidence scores to human override rates; recalibrate thresholds if auto-approved refills show >2% pharmacist reversal rate
- **Safety signal detection:** Identify patterns in flagged interactions that correlate with adverse events or prescriber intervention
- **Adherence modeling:** Refine early-refill risk scoring using actual pickup behavior and patient-reported reasons
- **Prompt optimization:** A/B test prompt variations on reasoning clarity and decision accuracy
- **Knowledge base updates:** Incorporate new clinical guidelines, formulary changes, and regulatory updates

Simulation 5 proposed: Replay historical refill requests through updated models to measure precision/recall on escalation decisions before deploying changes.

Continuous Learning: Human override rationales feed back into few-shot prompt examples and fine-tuning datasets proposed.

Governance Controls Applied to Healthcare Domain

Critical Control Points 1 5:

1. **Controlled Substances:** Deterministic DEA schedule checks and refill limits enforced in Stage 2 and Stage 6; zero tolerance for LLM hallucination on regulatory compliance proposed
2. **Drug Interactions:** Safety flags from Stage 2 force pharmacist review in Stage 6 regardless of confidence proposed
3. **Patient Safety:** Any adverse event flag in patient history triggers human review path proposed
4. **Audit Trail:** Full decision lineage required for regulatory inspections and malpractice defense 5 proposed

Regulatory Considerations: Compliance with HIPAA (data access controls), state pharmacy boards (refill limits), DEA (controlled substances), and insurance contracts (prior authorization) must be validated with client legal team proposed.

Assumptions

- Pharmacy and EHR systems expose APIs or HL7/FHIR interfaces for signal ingestion and execution

- Patient consent obtained for AI-assisted refill decisions under HIPAA
- Prescriber standing orders permit pharmacy-initiated refills for maintenance medications
- Insurance formulary data available in near-real-time
- 10-30 second decision latency acceptable for standard refills; emergency medications use deterministic fast-path
- Pharmacist and prescriber capacity exists to handle escalated reviews (estimated 20-30% of requests)
- LLM reasoning does not constitute medical diagnosis or prescribing (decision support only)

What to Validate with the Client

- **Regulatory jurisdiction:** State-specific pharmacy laws on refill automation, controlled substance handling, and pharmacist-in-charge oversight requirements
- **Prescriber engagement model:** Which providers have opted into automated refill protocols; how prescriber consults are routed and SLA expectations
- **Insurance integration:** Real-time formulary lookup feasibility; prior authorization workflow integration
- **Patient communication preferences:** Notification channels, language support, accessibility requirements
- **Clinical safety governance:** Which medication classes require mandatory pharmacist review; adverse event reporting workflows
- **Audit and compliance:** Retention periods for decision logs; state board of pharmacy inspection readiness
- **Emergency refill criteria:** Which medications qualify for fast-path approval; after-hours escalation procedures
- **Human review capacity:** Current pharmacist workload; feasibility of handling projected escalation volume
- **Liability and malpractice coverage:** Legal review of AI-assisted decision framework and informed consent language

SOURCES

BMRS_Solution_Architecture.docx https://ank-aws-learning-144160833556.s3.ap-southeast-2.amazonaws.com/ey-fde/BMRS_Solution_Architecture.docx

Enterprise_AI_Automation_Framework.docx https://ank-aws-learning-144160833556.s3.ap-southeast-2.amazonaws.com/ey-fde/Enterprise_AI_Automation_Framework.docx

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TRACE

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